

Localized Spectral Inductive Bias for Sequential Recommendation with Fixed Wavelet Packets

Anonymous submission

Abstract

Frequency-domain sequential recommendation has recently attracted increasing attention by viewing user behavior sequences as temporal signals composed of different spectral components. Existing studies demonstrate that spectral modeling can capture long-term preferences and short-term interest fluctuations that are difficult to characterize directly in the time domain. However, Fourier-based methods operate on global sinusoidal bases and may blur localized non-stationary behavior patterns, while wavelet-enhanced and adaptive filtering models often introduce learnable filters or dynamic frequency modules, adding trainable parameters and structural complexity. To address these limitations, we propose StaRec, a stability-aware sequential recommendation model with a fixed wavelet packet inductive bias. StaRec decomposes hidden item representations using non-learnable orthogonal wavelet packet filters and keeps the transform kernels fixed throughout training. Lightweight coefficient and channel scaling is applied to selected packet components for task-specific adaptation, after which all terminal packets participate in inverse reconstruction. The reconstructed spectral features are fused with a causal self-attention branch to jointly model localized multi-scale structures and contextual item transitions. Extensive experiments on four benchmark datasets demonstrate that StaRec outperforms six representative baselines on standard top- K ranking metrics. Further ablation and sensitivity studies demonstrate branch complementarity and robustness across wavelet configurations.

1 Introduction

Sequential recommendation (SR) predicts the next item by modeling a user’s chronologically ordered behavior history. Existing studies have developed a broad range of sequence encoders, including Markov-chain and recurrent models (Rendle, Freudenthaler, and Schmidt-Thieme 2010; Hidasi et al. 2015), attention-based and Transformer architectures (Vaswani et al. 2017; Kang and McAuley 2018; Sun et al. 2019; Li, Wang, and McAuley 2020), convolutional, memory-based, and graph-based models (Li et al. 2017; Liu et al. 2018; Tang and Wang 2018; Yuan et al. 2019; Wu et al. 2019; Ren et al. 2019), and self-supervised or contrastive

learning methods (Zhou et al. 2020; Xie et al. 2022; Qiu et al. 2022; Chen et al. 2022). Recently, frequency-domain SR has attracted increasing attention because user behavior can be viewed as a temporal signal containing components at different frequencies. Slowly varying components characterize persistent preference tendencies, whereas rapidly varying components capture abrupt short-term interests and local behavioral transitions. This perspective enables frequency-aware models to extract preference patterns that are difficult to describe directly through time-domain dependencies alone.

Existing frequency-domain recommenders mainly follow two technical routes. The first performs spectral modeling in global Fourier coordinates. FMLPRec replaces self-attention with filter-enhanced MLP layers and conducts token mixing in the frequency domain (Zhou et al. 2022). FEAREC introduces frequency-enhanced attention to improve sequential representation learning (Du et al. 2023b). BSAREC further analyzes the low-pass tendency of self-attention and injects a Fourier-domain inductive bias to balance low- and high-frequency information (Shin et al. 2023). These methods establish the effectiveness of spectral modeling for SR. However, Fourier bases consist of global sinusoids, and every transformed coefficient depends on the complete sequence. Such representations effectively capture sequence-wide frequency responses, but can obscure non-stationary local fluctuations, especially when a user’s intent changes within a short subsequence.

The second route focuses on localized or adaptive frequency modeling. Recent methods employ heterogeneous frequency experts, user-conditioned filters, and wavelet-enhanced transformations to improve personalization and recover local temporal variation (Zhang et al. 2025; Baek et al. 2025; Xu et al. 2026). WEAREC, for example, combines dynamic frequency filtering with wavelet feature enhancement to preserve personalized global-frequency information while recovering local non-stationary patterns (Xu et al. 2026). These designs substantially expand the range of spectral responses available to a recommender. At the same time, stronger adaptation requires more components of the frequency response to be estimated from interaction data. This creates a central design tension in frequency-domain SR: global Fourier models provide simple and effective spectral views but lack temporal localization, whereas highly adaptive

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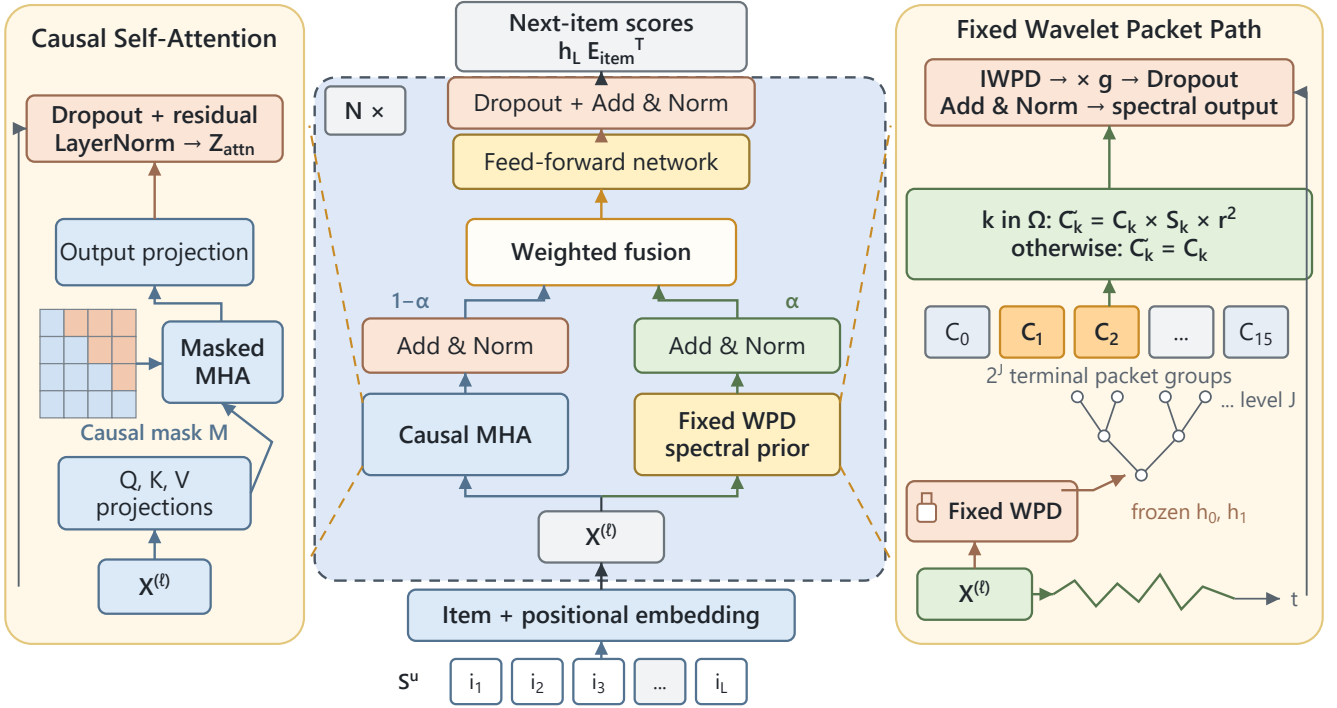


Figure 1: Overall architecture of StaRec. Item and positional embeddings are processed by N encoder blocks. In each block, causal self-attention models contextual item transitions under a future mask, while the fixed WPD path decomposes each hidden channel into 2^J terminal packet groups. Selected groups $k \in \Omega$ receive coefficient and channel scaling, unselected groups remain unchanged, and all groups participate in inverse reconstruction. The two residual branch outputs are combined by weighted fusion, followed by a feed-forward network and shared-embedding next-item prediction.

frequency models improve flexibility and locality at the cost of additional estimation burden and structural complexity.

To address this tension, we propose *StaRec*, a sequential recommendation model built on fixed wavelet packet priors. StaRec treats hidden item representations as temporal signals and decomposes them using predefined orthogonal wavelet packet filters. Unlike fully learnable spectral transforms, its analysis and reconstruction filters remain fixed throughout optimization and provide a stable localized coordinate system. Recommendation-specific adaptation is restricted to coefficient and channel scaling for selected terminal packet groups. Importantly, packet selection determines which groups are adapted rather than which groups are retained: all terminal packets participate in inverse wavelet packet reconstruction. The reconstructed spectral features are then fused with a parallel causal self-attention branch, allowing StaRec to combine localized multi-scale structure with adaptive item-to-item dependency modeling.

Wavelet packet decomposition (WPD) is particularly suitable for this design because it recursively decomposes both low- and high-pass branches, producing a full binary tree of localized subbands (Mallat 2002; Coifman and Wickerhauser 1992). This finer full-tree representation enables the spectral branch to describe both slowly varying preference tendencies and rapidly changing local behavior within a unified transform. By separating fixed coordinate construction from task-

specific coefficient adaptation, StaRec reduces the number of spectral components whose structure must be learned directly from recommendation data. Under the ideal orthogonal WPD formulation, the transform basis contributes no additional spectral-norm amplification to the core reconstruction operator; its sensitivity is governed by the learned coefficient and channel scaling operators. This property provides the structural foundation for the fixed-basis design.

The main contributions of this work are summarized as follows:

- We propose a fixed full-tree wavelet packet inductive bias for sequential recommendation. StaRec integrates localized multi-scale WPD coordinates with causal self-attention, combining structured spectral modeling with adaptive contextual dependency learning.
- We design a constrained packet adaptation mechanism that learns coefficient and channel scaling for selected terminal packet groups while retaining all packets during inverse reconstruction. We further characterize the resulting ideal orthogonal core operator and show how fixing the transform basis constrains its sensitivity.
- We conduct extensive experiments against six representative baselines on four benchmark datasets under a common full-sort protocol. StaRec achieves the strongest observed performance across all reported ranking metrics, while ablation, sensitivity, and efficiency analyses

demonstrate branch complementarity, dataset-dependent wavelet behavior, and the computational cost of reconstruction.

2 Preliminaries

2.1 Problem Formulation

Let $\mathcal{U}$ and $\mathcal{I}$ denote the user and item sets. For user u , the chronological history is $S^u = [i_1^u, \dots, i_{|S^u|}^u]$. Given a prefix $S_{1:t}^u$, sequential recommendation ranks all candidate items to predict i_{t+1}^u . Following standard settings (Kang and McAuley 2018; Sun et al. 2019), sequences are truncated or padded to length L , and padding positions are masked. Hidden sequences are represented by $\mathbf{X} \in \mathbb{R}^{L \times d}$.

2.2 Causal Self-Attention

Causal attention models content-dependent item transitions (Vaswani et al. 2017; Kang and McAuley 2018). For one attention head,

$$\begin{aligned} \mathbf{Q} &= \mathbf{XW}^Q, \quad \mathbf{K} = \mathbf{XW}^K, \quad \mathbf{V} = \mathbf{XW}^V, \\ \text{Attn}(\mathbf{X}, \mathbf{M}) &= \text{softmax}\left(\frac{\mathbf{QK}^\top}{\sqrt{d_k}} + \mathbf{M}\right) \mathbf{V}, \end{aligned} \quad (1)$$

where $\mathbf{M}$ masks future interactions. Multi-head attention applies this operation in parallel representation subspaces.

2.3 Wavelet Packet Decomposition

Wavelets use finite filters localized in time and scale (Mallat 2002; Daubechies 1990). Let $\mathbf{C}_\emptyset = \mathbf{x}$ and let p denote a path in the packet tree. A depth- J WPD recursively expands both low- and high-pass children and reconstructs from all terminal packets:

$$\begin{aligned} \mathbf{C}_{p0} &= \downarrow_2 (\mathbf{C}_p * h_0), \quad \mathbf{C}_{p1} = \downarrow_2 (\mathbf{C}_p * h_1), \\ \{\mathbf{C}_k\}_{k=0}^{2^J-1} &= \text{WPD}_{J,w}(\mathbf{x}), \quad \mathbf{x} = \text{IWPD}_{J,w}(\{\mathbf{C}_k\}_{k=0}^{2^J-1}). \end{aligned} \quad (2)$$

Here, w is the wavelet family and each terminal group corresponds to one binary path (Coifman and Wickerhauser 1992). Unlike DWT, which recursively expands only the approximation branch, WPD forms a full tree of localized subbands.

3 Motivation

Causal self-attention constructs content-dependent relationships among observed items and is effective at modeling contextual transitions and long-range dependencies. However, attention does not explicitly organize hidden trajectories according to temporal scale. Consequently, persistent preferences, gradual behavioral changes, and abrupt short-term interests must all be inferred from the same attention representation. Introducing an explicit spectral coordinate system provides a complementary view in which variations at different temporal scales can be represented separately.

Fourier-based models offer such a spectral perspective. For a length- L sequence $\mathbf{x}$, the k -th discrete Fourier coefficient is

$$\hat{x}_k = \sum_{t=0}^{L-1} x_t \exp(-i 2\pi kt/L). \quad (3)$$

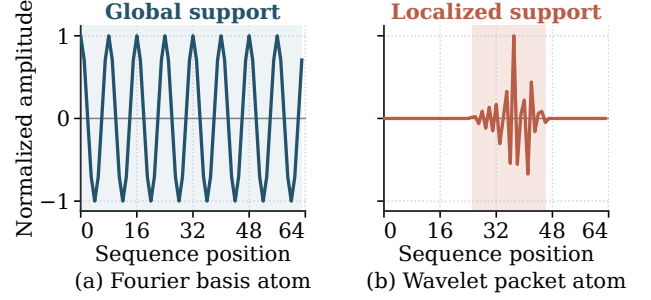


Figure 2: Support of representative normalized basis atoms. The length-64 Fourier atom has global temporal support, whereas the depth-3 db2 wavelet packet atom reconstructed from one terminal coefficient is localized within a shorter interval.

Because every temporal position contributes to every coefficient, Fourier bases effectively characterize sequence-wide periodic patterns but do not directly localize where a transient behavior occurs. As illustrated in Figure 2, a Fourier basis atom spans the complete sequence, whereas a wavelet packet atom is concentrated within a limited temporal region.

This localization is well suited to user histories containing both stable preferences and short-lived intent shifts. Moreover, WPD recursively decomposes both low- and high-pass branches, producing a full tree of localized subbands that describes smooth tendencies and rapidly varying behavior within a unified representation. StaRec therefore adopts fixed orthogonal WPD filters as its multi-scale spectral coordinates. Keeping the transform basis fixed separates coordinate construction from recommendation-specific response learning: selected packet coefficients and hidden channels are adapted through lightweight scaling, while all terminal packets participate in inverse reconstruction. A parallel causal self-attention branch remains responsible for content-dependent item transitions. The two branches consequently provide complementary structural and contextual views of the same interaction sequence.

4 Proposed Method

StaRec consists of N encoder blocks, each combining a fixed wavelet packet spectral branch, causal multi-head self-attention, and a position-wise feed-forward network, as illustrated in Figure 1. The spectral branch supplies localized multi-scale coordinates, while coefficient scaling and self-attention provide recommendation-specific adaptation.

4.1 Embedding Layer

For an interaction sequence S^u , item and positional embeddings are summed, normalized, and regularized by dropout:

$$\mathbf{X}^{(0)} = \text{Dropout}(\text{LN}(\mathbf{E}_{\text{item}}(S^u) + \mathbf{E}_{\text{pos}})), \quad (4)$$

where $\mathbf{X}^{(0)} \in \mathbb{R}^{L \times d}$, L is the maximum sequence length, and d is the hidden dimension.

Datasets	Metric	GRU4Rec	SASRec	BERT4Rec	FMLPRec	BSARec	WEARec	StaRec	Improv.
ML-1M	HR@5	0.1005	0.1374	0.1485	0.1331	0.1896	<u>0.2083</u>	0.2205	5.86%
	HR@10	0.1657	0.2137	0.2336	0.2093	0.2791	<u>0.2945</u>	0.3088	4.86%
	NG@5	0.0619	0.0873	0.0983	0.0861	0.1275	<u>0.1414</u>	0.1504	6.36%
	NG@10	0.0828	0.1116	0.1256	0.1105	0.1562	<u>0.1692</u>	0.1788	5.67%
Digital	HR@5	0.0108	0.0570	0.0612	0.0538	0.0837	<u>0.0838</u>	0.0870	3.82%
	HR@10	0.0191	0.0770	0.0855	0.0783	0.1086	<u>0.1093</u>	0.1117	2.20%
	NG@5	0.0066	0.0411	0.0453	0.0390	<u>0.0638</u>	<u>0.0632</u>	0.0663	3.92%
	NG@10	0.0092	0.0476	0.0531	0.0469	<u>0.0718</u>	0.0714	0.0743	3.48%
Musical	HR@5	0.0265	0.0458	0.0638	0.0513	0.0734	<u>0.0749</u>	0.0766	2.27%
	HR@10	0.0440	0.0634	0.0836	0.0669	<u>0.0953</u>	<u>0.0949</u>	0.0960	0.73%
	NG@5	0.0177	0.0335	0.0498	0.0395	0.0599	<u>0.0613</u>	0.0623	1.63%
	NG@10	0.0232	0.0392	0.0562	0.0469	0.0670	<u>0.0677</u>	0.0685	1.18%
Office	HR@5	0.0302	0.0548	0.0733	0.0670	0.0866	<u>0.0885</u>	0.0936	5.76%
	HR@10	0.0407	0.0677	0.0860	0.0796	0.1017	<u>0.1029</u>	0.1070	3.98%
	NG@5	0.0220	0.0444	0.0621	0.0552	0.0742	<u>0.0760</u>	0.0807	6.18%
	NG@10	0.0254	0.0485	0.0662	0.0592	0.0791	<u>0.0806</u>	0.0850	5.46%

Table 1: Overall full-sort recommendation performance. Bold and underline denote the best and second-best results, respectively. NG abbreviates NDCG. Improv. is the relative improvement of StaRec over the strongest baseline in each row.

4.2 Fixed Wavelet Packet Spectral Branch

Given the input $\mathbf{X}^{(\ell)}$ to block ℓ , WPD is applied independently to each hidden channel along the sequence dimension:

$$\left\{ \mathbf{C}_k^{(\ell)} \right\}_{k=0}^{2^J-1} = \text{WPD}_{J,w}(\mathbf{X}^{(\ell)}), \quad (5)$$

where w is the selected orthogonal wavelet family and J is the decomposition depth. The analysis and synthesis filters are stored as non-trainable buffers. A depth- J decomposition produces 2^J terminal packet groups, each corresponding to one path in the full binary packet tree.

Let $\Omega \subseteq \{1, \dots, 2^J - 1\}$ denote the terminal groups selected for learnable adaptation. The reported configuration uses $\Omega = \{1, \dots, \min(J, 2^J - 1)\}$ according to the terminal traversal order returned by the WPD module. For $\mathbf{C}_k^{(\ell)}, \mathbf{S}_k \in \mathbb{R}^{L_k \times d}$, the adapted coefficients are

$$\tilde{\mathbf{C}}_k^{(\ell)} = \begin{cases} \mathbf{C}_k^{(\ell)} \odot \mathbf{S}_k \odot \mathbf{r}^{\odot 2}, & k \in \Omega, \\ \mathbf{C}_k^{(\ell)}, & k \notin \Omega, \end{cases} \quad (6)$$

where $\mathbf{r} \in \mathbb{R}^d$ is broadcast over coefficient positions. The squared vector $\mathbf{r}^{\odot 2}$ ensures non-negative channel modulation, while the unconstrained $\mathbf{S}_k$ retains signed coefficient responses. Packet selection determines which groups are adapted; every terminal group remains in reconstruction.

All terminal packets are passed to inverse WPD, followed by output channel scaling and a residual Post-LN connection:

$$\begin{aligned} \hat{\mathbf{X}}^{(\ell)} &= \text{IWPD}_{J,w}(\{\tilde{\mathbf{C}}_k^{(\ell)}\}_{k=0}^{2^J-1}) \odot \mathbf{g}, \\ \mathbf{Z}_{\text{spec}}^{(\ell)} &= \text{LN}(\mathbf{X}^{(\ell)} + \text{Dropout}(\hat{\mathbf{X}}^{(\ell)})), \end{aligned} \quad (7)$$

where $\mathbf{g} \in \mathbb{R}^d$ is broadcast along the sequence dimension.

4.3 Fixed-Basis Operator Analysis

Let $\mathbf{W}_J \in \mathbb{R}^{L \times L}$ represent a depth- J orthogonal WPD for one channel. Under channel-major vectorization, its d -channel extension is $\mathcal{W}_J = \mathbf{I}_d \otimes \mathbf{W}_J$. Let $\mathcal{D}$ be the diagonal

coefficient-response operator induced by Equation 6, and let $\mathcal{G}$ repeat the output scales g_c over temporal positions. The ideal core reconstruction has the form

$$\begin{aligned} \mathcal{W}_J^\top \mathcal{W}_J &= \mathbf{I}_{dL}, \quad \|\mathcal{W}_J\|_2 = \|\mathcal{W}_J^\top\|_2 = 1, \\ \text{vec}(\mathcal{T}(\mathbf{X})) &= \mathcal{G} \mathcal{W}_J^\top \mathcal{D} \mathcal{W}_J \text{vec}(\mathbf{X}). \end{aligned} \quad (8)$$

Proposition 1. For fixed $\mathcal{D}$ and $\mathcal{G}$ and any $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{L \times d}$,

$$\|\mathcal{T}(\mathbf{X}) - \mathcal{T}(\mathbf{Y})\|_F \leq \|\mathcal{G}\|_2 \|\mathcal{D}\|_2 \|\mathbf{X} - \mathbf{Y}\|_F. \quad (9)$$

Indeed, Equation 8, norm equivalence under vectorization, and submultiplicativity give the product of the four operator norms. Orthogonality sets both WPD factors to one, yielding Equation 9. Because $\mathcal{D}$ and $\mathcal{G}$ are diagonal, $\|\mathcal{D}\|_2 = \max\{1, \max_{k \in \Omega, i, c} |S_{k,i,c} r_c^2|\}$ and $\|\mathcal{G}\|_2 = \|\mathbf{g}\|_\infty$. Thus the fixed orthogonal basis introduces no additional spectral-norm factor; the complete proof is provided in the supplementary material.

4.4 Parameter and Computational Complexity

Let q be the wavelet filter length and L_k the coefficient length of selected group k . For one encoder block,

$$\begin{aligned} |\Theta_{\text{spec}}| &= d \sum_{k \in \Omega} L_k + 2d, \\ \text{Cost}_{\text{WPD+IWPD}} &= O(JLdq). \end{aligned} \quad (10)$$

The first term counts the coefficient matrices and the two channel-scaling vectors. The fixed analysis and synthesis filters add no trainable parameters. Since aggregate packet length is $O(L)$ at each level, analysis and reconstruction are linear in sequence length for fixed J and q .

4.5 Attention Fusion and Prediction

Let $\text{MHA}(\mathbf{X}^{(\ell)}, \mathbf{M})$ denote causal multi-head attention, where $\mathbf{M}$ masks future interactions and padding positions.

The attention output, weighted fusion, and feed-forward update are

$$\begin{aligned} \mathbf{Z}_{\text{attn}}^{(\ell)} &= \text{LN}\left(\mathbf{X}^{(\ell)} + \text{Dropout}(\text{MHA}(\mathbf{X}^{(\ell)}, \mathbf{M}))\right), \\ \mathbf{Z}^{(\ell)} &= \alpha \mathbf{Z}_{\text{spec}}^{(\ell)} + (1 - \alpha) \mathbf{Z}_{\text{attn}}^{(\ell)}, \quad \alpha \in [0, 1], \\ \mathbf{X}^{(\ell+1)} &= \text{LN}\left(\mathbf{Z}^{(\ell)} + \text{Dropout}(\text{FFN}(\mathbf{Z}^{(\ell)}))\right). \end{aligned} \quad (11)$$

The fusion weight α balances localized spectral structure and contextual item dependencies.

Both branches apply a residual connection and LayerNorm before fusion, placing their outputs on comparable scales. StaRec treats α as a dataset-level hyperparameter selected on validation data rather than introducing an additional input-dependent gating network. The resulting convex combination preserves a lightweight fusion path: $\alpha = 0$ recovers the attention branch, $\alpha = 1$ isolates the spectral branch, and intermediate values allow localized multi-scale features to complement contextual transitions.

After the final block, the last valid state $\mathbf{h}_L$ scores all candidates through the shared item embedding matrix. For the ground-truth next item y , the training objective is

$$\mathbf{o} = \mathbf{h}_L \mathbf{E}_{item}^\top, \quad \mathcal{L} = -\log \frac{\exp(o_y)}{\sum_{i \in \mathcal{I}} \exp(o_i)}. \quad (12)$$

5 Experiments

We evaluate StaRec by answering the following research questions:

- **RQ1:** How does StaRec compare with representative sequential recommendation models, particularly frequency-domain methods?
- **RQ2:** Do the fixed WPD branch and causal self-attention provide complementary information?
- **RQ3:** How sensitive is StaRec to the wavelet family and decomposition depth?
- **RQ4:** What parameter and inference-time overhead does the WPD branch introduce?

5.1 Experimental Setup

Datasets. We conduct experiments on MovieLens-1M (ML-1M) (Harper and Konstan 2015) and three categories from the Amazon review collection (McAuley et al. 2015): Digital Music, Musical Instruments, and Office Products. ML-1M is a dense benchmark with long interaction histories, whereas the Amazon categories contain shorter and substantially sparser sequences. These datasets cover different domains, catalog sizes, and sequence-length distributions, allowing us to examine StaRec under heterogeneous recommendation settings.

Baselines. We compare StaRec with six representative methods. GRU4Rec (Hidasi et al. 2015) models user behavior with a recurrent encoder. SASRec (Kang and McAuley 2018) applies causal self-attention, while BERT4Rec (Sun et al. 2019) uses bidirectional Transformer encoding and masked-item prediction. FMLPRec (Zhou et al. 2022) performs token mixing through learnable filtering in Fourier

coordinates. BSARec (Shin et al. 2023) combines self-attention with Fourier-based low- and high-frequency rescaling. WEARec (Xu et al. 2026) integrates dynamic frequency filtering with wavelet feature enhancement. Together, these baselines cover recurrent, attention-based, MLP-filtering, Fourier-bias, and adaptive wavelet-enhanced architectures.

Evaluation protocol. For every user, the last interaction is used for testing, the penultimate interaction for validation, and all preceding interactions for training. All models use the same processed sequences, maximum sequence length, leave-one-out split, history masking, and full-sort candidate set. We report HR and NDCG at cutoffs 5 and 10. Validation NDCG@10 is used for checkpoint selection, hyperparameter selection, and early stopping.

Implementation details. StaRec is implemented in PyTorch with maximum sequence length 50, hidden size 64, two encoder blocks, two attention heads, and batch size 256. We optimize full-vocabulary cross-entropy using Adam with $(\beta_1, \beta_2) = (0.9, 0.999)$, zero weight decay, at most 200 epochs, and early-stopping patience 10. We perform a Cartesian grid search over learning rate $\{10^{-4}, 3 \times 10^{-4}, 3 \times 10^{-3}\}$, dropout $\{0.1, 0.3, 0.5\}$, fusion weight $\alpha \in \{0.1, 0.2, 0.3, 0.5, 0.8\}$, and wavelet family $\{\text{haar}, \text{db2}, \text{db4}, \text{sym4}, \text{sym6}\}$, giving 225 candidate configurations per dataset. The final settings are selected by validation NDCG@10 and are listed in the supplementary material.

Experiments are executed on a Linux server with two Intel Xeon Gold 6348 CPUs, 1 TiB RAM, and eight NVIDIA A800-SXM4-80GB GPUs, with one GPU used for each run. The software environment consists of Ubuntu 22.04.4 LTS, Python 3.10.12, CUDA 13.0, PyTorch 2.12.0, NumPy 2.2.6, SciPy 1.15.3, and PyWavelets 1.8.0.

5.2 Overall Performance (RQ1)

Table 1 reports the full-sort recommendation performance. StaRec obtains the best result for all four metrics on all four datasets. Relative to the strongest baseline in each row, the improvement ranges from 0.73% to 6.36%, demonstrating consistent gains across dense long-history and sparse short-history recommendation settings.

On ML-1M, StaRec improves over WEARec by 4.86–6.36%, with the largest gain on NDCG@5. Office Products shows similarly consistent improvements of 3.98–6.18%. Gains on Digital Music and Musical Instruments are smaller but remain positive for every metric. The results demonstrate that fixed WPD features complement attention across both long-history and short-history datasets.

5.3 Branch Ablation (RQ2)

We compare StaRec with two variants. *w/o Spectral* disables the fixed WPD branch while retaining causal self-attention, and *w/o Attention* retains the spectral branch while removing self-attention. ML-1M and Office Products represent dense long-history and sparse short-history recommendation settings, respectively.

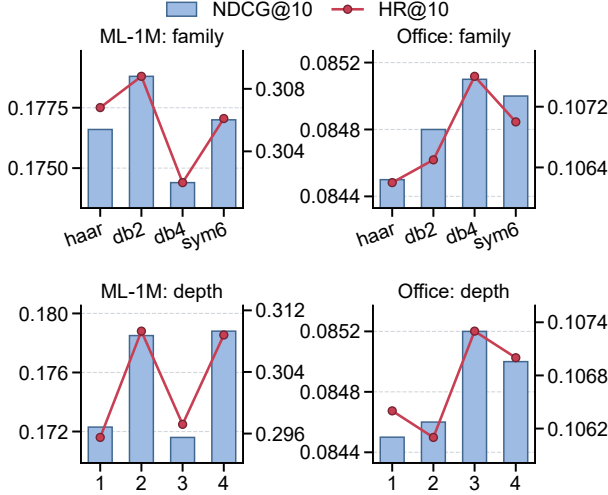


Figure 3: Sensitivity of StaRec to wavelet family and decomposition depth on ML-1M and Office Products. Bars report NDCG@10 and lines report HR@10.

Methods	ML-1M		Office Products	
	HR@10	NDCG@10	HR@10	NDCG@10
StaRec	.3088	.1788	.1070	.0850
w/o Spectral	.2992	.1740	.1061	.0841
w/o Attention	.2740	.1594	.0952	.0755

Table 2: Branch ablation on ML-1M and Office Products. The best result in each column is bolded.

Removing the spectral branch reduces HR@10 and NDCG@10 by 3.11% and 2.68% on ML-1M, and by 0.84% and 1.06% on Office Products. Removing attention causes substantially larger reductions of 10.85–11.27% on ML-1M and 11.03–11.18% on Office Products. These results confirm the intended division of labor: fixed WPD supplies complementary multi-scale information, while attention remains the principal contextual sequence model.

5.4 Hyperparameter Sensitivity (RQ3)

Figure 3 studies the effects of wavelet family and decomposition depth on ML-1M and Office Products. For wavelet selection, ML-1M performs best with db2, reaching .3088 HR@10 and .1788 NDCG@10, whereas Office Products reaches its highest values with db4. The differences among Haar, db2, db4, and sym6 remain modest, particularly on Office Products.

For decomposition depth, ML-1M obtains its highest HR@10 at depth 2 and its highest NDCG@10 at depth 4. Office Products performs best at depth 3 for both metrics, with depth 4 remaining close. The results show that the preferred wavelet configuration depends on the interaction regime, while performance remains stable across neighboring choices.

5.5 Parameter and Runtime Analysis (RQ4)

Table 3 reports parameter count and wall-clock time for one full-sort evaluation pass on Office Products. StaRec introduces only a small parameter increase because its wavelet filters are fixed buffers and the trainable additions consist primarily of coefficient and channel scaling parameters.

Methods	Office Products	
	# params	s/eval
StaRec	1,874,368	60.5252
SASRec	1,863,360	51.1587
FMLPRec	1,836,736	48.4208
BSARec	1,863,744	54.1214

Table 3: Parameter count and full-sort evaluation time on Office Products. s/eval denotes seconds per full-sort evaluation pass.

StaRec uses 0.59% more parameters than SASRec and 0.57% more than BSARec. Its evaluation time is 18.31% higher than SASRec and 11.83% higher than BSARec because each encoder block performs WPD and inverse reconstruction in addition to attention. The additional computation yields the accuracy improvements reported in Table 1 while keeping the total parameter growth below one percent.

5.6 Summary of Findings

The experiments provide four observations. First, StaRec achieves the best result on every evaluated dataset and metric. Second, the WPD and attention branches are complementary, with attention contributing the stronger standalone sequence representation. Third, wavelet family and decomposition depth have dataset-dependent effects, confirming the importance of validation-based configuration. Finally, the fixed WPD branch adds less than one percent trainable parameters and introduces a moderate inference-time overhead through decomposition and reconstruction.

6 Related Work

6.1 Sequential Recommendation

Neural sequential recommendation has evolved from recurrent and convolutional encoders (Hidasi et al. 2015; Yuan et al. 2019) to graph-based session models (Wu et al. 2019) and Transformer architectures. SASRec applies causal self-attention for next-item prediction (Kang and McAuley 2018); BERT4Rec adopts bidirectional masked-item prediction (Sun et al. 2019); TiSASRec explicitly incorporates temporal interaction intervals (Li, Wang, and McAuley 2020); and LightSANs and AdaMCT modify attention structures to improve efficiency or adaptively mix local and global preferences (Fan et al. 2021; Jiang et al. 2023). At the system level, Transformers4Rec provides a flexible Transformer framework for sequential and session recommendation (de Souza Pereira Moreira et al. 2021), while RecBole supports unified algorithm implementation and evaluation (Zhao et al. 2020).

Another major line of work relies on self-supervised contrastive learning across sequence views: S3-Rec maximizes

mutual information among contextual entities (Zhou et al. 2020), CoSeRec and CL4SRec leverage behavior-aware or stochastic augmentations (Liu et al. 2021; Xie et al. 2022), DuoRec alleviates representation degeneration via semantic positive sampling (Qiu et al. 2022), ICLRec incorporates latent user intents (Chen et al. 2022), and CauseRec synthesizes counterfactual sequences to isolate causal preferences (Zhang et al. 2021). These methods primarily optimize auxiliary training objectives or data augmentation strategies. In contrast, StaRec directly alters the encoder’s structural spectral prior by integrating localized multi-scale coordinates into causal attention.

6.2 Frequency-Domain and Wavelet Recommendation

Predefined spectral transforms can effectively serve as parameter-free token mixers, as demonstrated by FNet outside recommendation (Lee-Thorp et al. 2022). In sequential recommendation, Fourier-based methods model global frequency characteristics: FMLPRec learns frequency-domain filters directly in Fourier coordinates (Zhou et al. 2022); SLIME4Rec couples frequency-aware mixing with contrastive objectives (Du et al. 2023a); FEAREC combines time- and frequency-domain attention mechanisms (Du et al. 2023b); and BSAREC adds explicit low- and high-frequency Fourier biases to self-attention (Shin et al. 2023). Subsequent extensions further enhance spectral adaptability through heterogeneous experts, user-conditioned filters, or inter-session frequency relations (Zhang et al. 2025; Baek et al. 2025; He et al. 2026).

Wavelet transforms overcome Fourier’s lack of temporal localization by providing multi-resolution basis functions localized in both time and frequency (Mallat 2002; Daubechies 1990), with wavelet packet decomposition (WPD) further partitioning both approximation and detail subbands into a full binary tree (Coifman and Wickerhauser 1992). Wavelet-based recommendation approaches utilize these coordinates differently: DWTRec constructs an adaptive DWT-based time-frequency filter (Lu et al. 2025), WEAREC combines dynamic Fourier filtering with wavelet feature enhancement (Xu et al. 2026), and WPGRec couples stationary wavelet packet transforms with graph propagation and gated subband fusion (Liu, Ji, and Yan 2026). StaRec occupies a distinct design point in this landscape: a decimated full-tree WPD basis remains fixed, selected terminal packets receive lightweight coefficient scaling, all terminal packets are preserved during reconstruction, and causal self-attention serves as the parallel adaptive dependency model. This parameterization separates localized spectral structure from content-dependent transition modeling while retaining complete packet reconstruction.

7 Conclusion

This paper investigates sequential recommendation from the perspective of localized spectral modeling. Although causal self-attention effectively captures content-dependent item transitions, it does not explicitly organize hidden sequences across localized temporal scales. Meanwhile, Fourier-based

representations rely on globally supported basis functions and may obscure short-lived behavioral variations. To address these limitations, we proposed StaRec, which combines causal self-attention with a fixed full-tree wavelet packet branch. The fixed orthogonal WPD basis provides localized multi-scale coordinates, while lightweight coefficient and channel scaling adapts selected packet components to the recommendation objective. Our operator analysis further shows that trainable spectral variation is concentrated in diagonal scaling operators rather than the transform basis. Experiments on four benchmark datasets demonstrate that StaRec outperforms six representative baselines on every reported HR and NDCG metric. Ablation studies confirm the complementarity of the spectral and attention branches, while sensitivity and efficiency analyses characterize the effects of wavelet configuration, parameter growth, and reconstruction cost. Future work will investigate adaptive packet selection, richer interactions across packet groups, and more efficient wavelet reconstruction for large-scale sequential recommendation.

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